higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL PRINCIPLES AND PRACTICE
NQF LEVEL 3

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This marking guideline consists of 7 pages.

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QUESTION 1

1.1 1.1.1 An atom consists of protons and neutrons in the nucleus and electrons orbiting around the nucleus. (3)

1.1.2 Atomic mass is the mass of the nucleus of an atom. (1)

1.2 As temperature increases, the resistance of a germanium semiconductor decreases. (2)

1.3
- Current is the flow of electric charge through a conductor.
- Electrons are particles with a negative charge that move around a nucleus. (2)

1.4 $R_t = R_0[1 + \propto_0 (t - \emptyset)]$
$= 10[1 + 0{,}0039(60 - 20)]$ ✓✓
$= 11{,}56\ \Omega$ ✓ (3)

1.5 1.5.1 True
1.5.2 False
1.5.3 True
1.5.4 False
(4 × 1) (4)
[15]

QUESTION 2

2.1
- Relay
- Bell
- Lifting magnet (3)

2.2 Magnetic field strength is a measure of the magnetic intensity of the field. (2)

2.3 $B = \frac{\emptyset}{A}$
$= \frac{800 \times 10^{-6}}{0{,}004}$ ✓
$= 0{,}2\ T$ ✓ (2)

2.4 2.4.1 $F = BIl\ sin\emptyset$
$= 0{,}45 \times 4 \times 0{,}15\ sin\ 90^0$ ✓
$= 0{,}27\ N$ ✓

2.4.2 $F = BIl\ sin\emptyset$
$= 0{,}3 \times 4 \times 0{,}15\ sin 60^0$ ✓
$= 0{,}156\ N$ ✓
(2 × 2) (4)

2.5 2.5.1 Current A (I)
2.5.2 Flux Wb (ϕ)
2.5.3 Resistance Ω (R)
2.5.4 Reluctance H^{-1}(S)

(4 × 1) (4)

2.6 If you hold a compass near a wire through which current is flowing, the needle on the compass will be deflected either clockwise or anticlockwise depending on the direction of the current. (3)

2.7
- Lenz's law
- Fleming's right-hand rule

(2)
[20]

QUESTION 3

3.1 3.1.1

$$R_S = R_1 + R_2$$
$$= 9 + 3$$
$$= 12\ \Omega \checkmark$$

$$R_p = \frac{R_S \times R_3}{R_S + R_3} \checkmark$$
$$= \frac{12 \times 6}{12 + 6}$$
$$= 4\ \Omega \checkmark$$

OR

$$\frac{1}{R_p} = \frac{1}{R_S} + \frac{1}{R_3}$$
$$= \frac{1}{12} + \frac{1}{6}$$
$$= \frac{3}{12}$$
$$R_p = \frac{12}{3} = 4\ \Omega$$

$$R_T = R_S + R_P$$
$$= 4 + 2$$
$$= 6\ \Omega \checkmark$$

(4)

3.1.2

$$I_T = \frac{V_T}{R_T}$$
$$= \frac{12}{6} \checkmark$$
$$= 2\ A \checkmark$$

(2)

3.1.3 $I_S = \frac{R_3}{R_3 + R_S} I_T$ ✓✓

$= \frac{6}{6+12} \times 2$

$I_S = I_{9\Omega} = 0{,}667\ A$ ✓✓

OR

$V_p = I_T \times R_p$

$= 2 \times 4$

$V_p = 8\ V$

$I_S = \frac{V_p}{R_S}$

$= \frac{8}{12}$

$I_S = I_{9\Omega} = 0{,}667\ A$ (4)

3.1.4 $R_T = R_3 + R_4$

$= 6 + 2$ ✓

$= 8\ \Omega$ ✓

$I_T = \frac{V_T}{R_T}$

$= \frac{12}{8}$ ✓

$= 1{,}5\ A$ ✓ (4)

3.2 3.2.1 The emf of cells is the greatest potential difference that can be produced by a source of electric current in a circuit.

3.2.2 Terminal voltage is the voltage measured across the terminals of the load in any system.

(2 × 2) (4)

3.3 $\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$

$= \frac{1}{20} + \frac{1}{4} + \frac{1}{5}$

$= \frac{10}{20}$ ✓

$L_T = \frac{20}{10}$ ✓

$= 2\ mH$ ✓ (3)

3.4 3.4.1

$$C_T = \frac{C_1 \times C_2}{C_1 + C_2}$$

$$= \frac{30 \times 10}{30 + 10} \checkmark$$

$$= 7{,}5\ \mu F \checkmark$$

OR

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$= \frac{1}{30} + \frac{1}{10}$$

$$= \frac{4}{30}$$

$$= \frac{30}{4}$$

$$C_T = 7{,}5\ \mu F$$

(2)

3.4.2

$$Q_T = C_T V_T$$
$$= 7{,}5 \times 10^{-6} \times 12 \checkmark$$
$$= 90\ \mu F \checkmark$$

$$V_2 = \frac{Q_2}{C_2}$$
$$= \frac{90}{10} \checkmark$$
$$= 9\ V \checkmark$$

(4)

3.5
- Electrolytic capacitor
- Tantalum capacitor

(2)

3.6 Step-down transformer (1)

[30]

QUESTION 4

4.1
- Uniform scale
- Low-power consumption
- Extendable range using series and shunt resistors

(Any 2 × 1) (2)

4.2 4.2.1 A damping device ensures that the pointer will not oscillate and that it will come to rest so that an accurate reading can be taken.

4.2.2 A controlling device ensures that the deflection shown on the meter is the measured quantity.

(2 × 2) (4)

4.3
- When current is passed through the solenoid, the two iron rods are magnetised in the same direction and therefore repel each other. This movement results in a deflection of the pointer.
- The fixed coil becomes an electromagnet as soon as current from the circuit flows through it and the coil magnetises the two iron discs. The two iron vanes will have the same magnetic polarity and will therefore repel each other. The deflecting torque produced by this force of repulsion causes the pointer to deflect away from zero. (Any 1 × 3) (3)

4.4 Parallax error occurs when the eye is not directly opposite the scale when taking a reading. (2)

4.5
$$R_{se} = \frac{V_T}{I_T} - R_m \qquad OR \qquad R_T = \frac{V_T}{I_T}$$
$$= \frac{10}{0{,}015} - 5 \checkmark\checkmark \qquad = \frac{10}{0{,}015} = 666{,}667\ \Omega$$
$$= 661{,}667\ \Omega \checkmark \qquad R_{se} = R_T - R_m = 666{,}667 - 5 = 661{,}667\ \Omega$$
(3)

4.6 The frequency meter is connected in parallel in a circuit. (1)
[15]

QUESTION 5

5.1
- Crane
- Hoist
- Conveyor (3)

5.2 5.2.1 DC machine (1)

5.2.2
A – Field winding
B – Pole shoe
C – Armature winding
D – Brush
E – Commutator (5)

5.3
- Armature
- Field winding
- Shaft
- Commutator
- Brushes (Any 4 × 1) (4)

5.4
$$I_{sh} = \frac{V}{R_{sh}} = \frac{250}{150} = 1{,}667\ A \checkmark$$
$$= I_L + I_{sh} = 55 + 1{,}667 = 56{,}667\ A \checkmark\checkmark$$
$$E = V + I_a R_a$$
$$E = 250 + (56{,}667 \times 0{,}05) \checkmark$$
$$E = 252{,}833\ V \checkmark$$
(5)

5.5
- High torque
- Woodworking machinery
- Air compressors (Any 2 × 1) (2)

[20]

TOTAL: 100

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